



Review

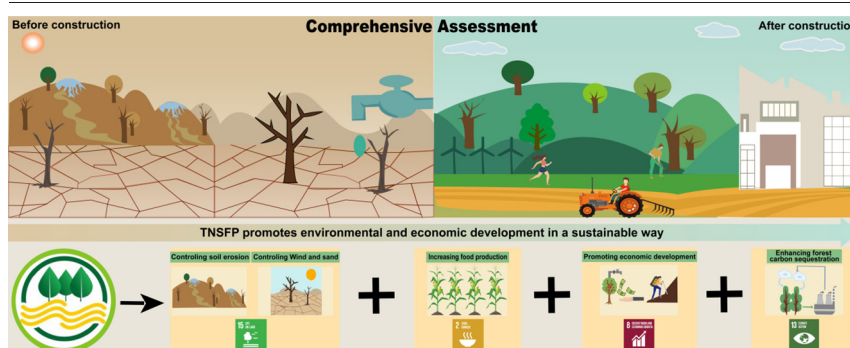
Assessing the effects of China's Three-North Shelter Forest Program over 40 years

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HIGHLIGHTS

- The ecological service and economic benefits of the TNSFP were reviewed.
- Clarify the relationship between the TNSFP to SDGs
- Summarized the lessons learned from the TNSFP construction
- Proposed the challenges and suggestions for the TNSFP construction

GRAPHICAL ABSTRACT



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ABSTRACT

Forests provide vital ecosystem services such as soil and water conservation, climate regulation, and carbon storage. Large-scale afforestation programs are being attempted in many countries to improve environmental conditions in deteriorated or unfavorable locations. China's Three Northern Protected Forest Program (TNSFP), accounting for 42.40 % of China's total land area, is the world's largest afforestation program to date. The TNSFP has continued providing critical ecosystem services to humans over 73 years (1978–2050) with a total investment of CNY 93.3 billion. To facilitate understanding of the TNSFP's contribution, the effects of the TNSFP for last 43 years were comprehensively evaluated by using integrated review of structured literature, bibliometric analysis, and thematic analysis. We incorporated and expanded the direct ecosystem services evidence of the TNSFP from wind and sand control, soil erosion control and carbon sequestration to indirect economic benefits, e.g. increasing crop yield and promoting economic development. We found that over the past 40 years of TNSFP construction, wind and sand hazards and soil erosion in China's Three-North areas have been effectively controlled, and forest carbon sequestration, grain production and economic output have steadily increased. The ecosystem services provided by the TNSFP are highly consistent with the thrust of the UN Sustainable Development Goals, and the TNSFP has contributed to the realization of SDG2, SDG8, SDG13, and SDG15. Although achieving tremendous ecological, economic, and social benefits, the TNSFP still has knowledge gaps in its scientific basis. And the limited local engagement and insufficient investment highly hinder the TNSFP from playing its multiple functions. We suggest several urgent actions and directions to address these limitations. This review could help researchers gain insight into key areas of ecological restoration in the TNSFP, providing a reference for future research in the TNSFP construction in China and other regions of the world embarking on similar journeys.

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1. Introduction

Environmental degradation is a global common concern to the international community because it causes a series of natural and social problems, such as natural disasters and food crisis (Zhou et al., 2021). Ecological restoration is the key element in rebuilding ecological equilibria and reversing environmental degradation (Hobbs et al., 2011; Ma et al., 2013; McKay et al., 2005). Large-scale afforestation programs are being attempted in many countries to improve environmental conditions in deteriorated or unfavorable locations (Zhu, 2013). However, the effects of these ecological restoration programs have sparked intense international debate (McTainsh et al., 2011). Some programs are not successfully completed as expected for socio-political, technical or ecological reasons, such as Great Plan for the Transformation of Nature in the former Soviet Union (Brain, 2010), Iran's National Action Program to Combat Desertification (Amiraslani and Dragovich, 2011), Israel's 22nd National Master Plan (Portnov and Safriel, 2004), and the Great Plains Protected Forest Program in the United States (Orth, 2007).

Since the 1970s, China has also undertaken a series of key forestry ecological development programs, including the Natural Forest Protection Program (NFPP), the Conversion of Cropland into Forests Program (CCFP), the Desertification Combating Program around Beijing and Tianjing (DCBT), the Key Shelterbelt Development Programs in the Three-North Region and in the Middle and Lower Reaches of the Yangtze River (KSDP) and the Nature Reserve Development Program in Forestry Sector (WCNR) (Wang et al., 2021). And China's Three Northern Protected Forest Program (TNSFP) is still under construction, which is the most ambitious reforestation Program of the world.

The TNSFP is planned to be completed in 2050, and involves three stages and seven phases, covering 725 counties and 13 provinces, with an area of 4.07×10^8 ha (42.40 % of China). It is the world's largest forestry ecological Program to date (Chu et al., 2019; Liu et al., 2009; Pan et al., 2011; Tan and Li, 2015; Zhao et al., 2017; Zhu et al., 2017). The TNSFP has been under construction for 43 years, with a total completed reforestation area of 30.14 million ha. Forest cover ranges from 5.05 % to 13.57 % in the program area. As an important initiative to use ecological restoration to combat ecological degradation, the TNSFP has improved the natural conditions of ecologically fragile areas and protected forest resources (Chu et al., 2019; Liu et al., 2009; Pan et al., 2011; Zhao et al., 2017). However, there is still no dedicated review on the ecological and economic benefits of the TNSFP, such as combating desertification and controlling dust storms,

controlling soil erosion, enhancing forest carbon sequestration, increasing crop yield, and promoting economic development. The linkage and the contribution of TNSFP to the UN Sustainable Development Goals (SDGs) have been neglected for a long time.

The Chinese government incorporated forestry into a “sustainable development” framework in the late 1990s. This new policy leads to the establishment of six key national forestry programs to protect the existing natural forests and restore the degraded ecosystems (NFPP, CCFP, DCBT, KSDP, WCNR) (Li, 2004; Xi et al., 2014). These programs have substantially contributed to the sustainable development of China (Liu and Diamond, 2005; Xi et al., 2012). However, to our knowledge, few comprehensive assessments have focused on the impact of TNSFP on China's achievement to the 2030 UN Sustainable Development Goals (SDGs), although the TNSFP has presented an noteworthy role in increasing food production and eliminating hunger (SDG2), promoting economic growth (SDG8), enhancing climate change mitigation (SDG13), and combating desertification and restoring degraded land and soil (SDG15). Thus, a comprehensive assessment of the TNSFP is urgently needed.

We comprehensively investigated the recent progress of the TNSFP by reviewing and synthesizing the relevant literature. Our objectives were to: (1) assess the latest achievements in ecological restoration and economic development for over 40 years of TNSFP construction; (2) assess the impacts of the TNSFP on sustainable development in the program area; (3) identify the challenges and limitations of the TNSFP; (4) propose future recommendations for the TNSFP from further scientific research to policy management to sustainable development. The knowledge gained from the comprehensive assessment of the TNSFP could provide experience for afforestation and reforestation, enhancing the role of forests in combating land degradation and desertification, coping with sand and dust storms, improving the ecological conservation benefits of forests (soil and water conservation, etc.), managing natural and planted forests in a sustainable way, and improving the socio-economic benefits of forests. Our work can also provide a reference for ecological restoration Programs around the world to achieve sustainable development.

2. Overview methodology

We used a combination of systematic literature review and thematic analysis to identify research themes and directions (Bastas and Liyanage, 2018; Kumar et al., 2022). First, within the context of our initial literature study, we assessed the breadth of current material on TNSFP and

determined that the research database contained a large body of potentially relevant literature. We searched for relevant research papers from the Google database. A set of keywords was defined, which are ("forest" or "forestry engineering" or "wind and sand control" or "soil erosion" or "ecological restoration") and ("sustainable development") and ("three northern protection forest Program"). When the phrase "Three North Protection Forest Program" was used as a search term, approximately 17,900 matches were initially found in this knowledge base. Our search timeframe will be limited to studies published or produced from 1979 (after the first year of TNSFP implementation) - 2021. The search language will be conducted in English and Chinese.

Then, literature screening based on inclusion and exclusion criteria was performed to identify papers to be read in detail. Including (subject area, "wind and sand control" or "ecological restoration" or "engineering" or "forestry" or "forestry" or "sustainable development"), we selected 341 articles, and excluding (document type "review"), 279 articles were retained. Then title and abstract screening will be performed for full text screening. Finally, we screened 98 articles being involved in at least one study metric to two or more metrics, depending on the number of metrics involved, thus performing integration to determine the final study metrics for the TNSFP. The research themes that emerge may not represent the entirety of the TNSFP field, and many of the key themes can only be further identified through detailed reading of papers and qualitative thematic analysis.

3. Changes of ecological environment in China's Three-North region

The natural geological and ecological patterns in China's Three-North region are highly complex. From 1100 to 1199, the population of the Horqin Sands in the western Northeast region gradually increased, agriculture and animal husbandry expanded, and substantial areas of forests were destroyed, leading to ecological degradation (Fig. 1). Between 1271 and 1368, an area of 3.00×10^4 ha of cultivated land became desertified in the northwest Alashan region because of human activities and natural factors. In the 1960s and 1970s, forest land conversion and agricultural land reclamation programs were implemented as part of the national strategy for food self-sufficiency, deforesting natural vegetation (Xu et al., 2006). During the same period, the ecological environment in Beijing, Yinchuan, Xining and Lanzhou in the Three-North region also deteriorated severely, and the annual average number of wind and sand days gradually increased.

The number of sand days in Beijing increased from 20.8 d to 22.3 d per year, while the number of sand days in Xining increased from 31.1 d to 57.8 d per year (Qian et al., 2002).

In 1978, the Chinese government proposed and implemented a reforestation program in the Three-North area to combat desertification and control dust storms and to control soil erosion—the Three-North Shelter Forest Program (Fig. 2) (Wang et al., 2004; Wang et al., 2010). The TNSFP can be divided into the construction of four shelterbelt systems in the western northeast, Mengxin, Loess Plateau, and northern China (Fig. 2) (Administration SFAG, 2019; Zhang et al., 2016; Zheng and Zhu, 2017; Zhu et al., 2017). Four silvicultural types are included in the TNSFP: semi-arid and semi-humid planted tree type species (e.g., poplar, elm spp., birch spp., oleander, camphor pine); semi-arid shrub species (tamarisk, brome, buckthorn); arid shrub-steppe species (*Artemisia* spp.); and extreme arid-unvegetated zone with no silvicultural strategy and with three plant sources: native species, hybrid species, and introduced species (Li et al., 2012; Zhu et al., 2017).

After the 1998 floods on the Yangtze River, which inundated 2.12×10^7 ha of land, the TNSFP-based forest conservation policies further contributed to the increase of forest cover in China (Hu et al., 2021; Viña et al., 2016; Zong and Chen, 2000). From 2001 to 2010 (Phase IV of the Program), the TNSFP had completed an afforestation area of 6.98×10^5 ha and saved a forest area of 4.43×10^6 ha (a save rate of 63.5 %). By 2005, the forest area increased to 2.9×10^7 ha. By 2008, the forest area of the TNSFP had increased further to 3.41×10^7 ha (Fig. 1) (Administration SFAG, 2019; Jia et al., 2015; Zhu et al., 2017). In 2017, the completed afforestation area cumulatively accounted for 118.2 % of the planned afforestation tasks for the same period in the Three-North region, and the forest coverage rate increased to 13.57 % with a general increase in vegetation cover (Duan et al., 2011; Li et al., 2012; Qiang et al., 2014). The TNSFP has invested a total of 93.3 billion CNY over 40 years (without considering the change in purchasing power of the currency, same below). The TNSFP has been the leading driver of this spatiotemporal change in China's Three-North region ecology.

4. Comprehensive assessment of the TNSFP

We collected evidence on the contribution of the TNSFP to sustainable development in five aspects from ecosystem service to economic benefits:

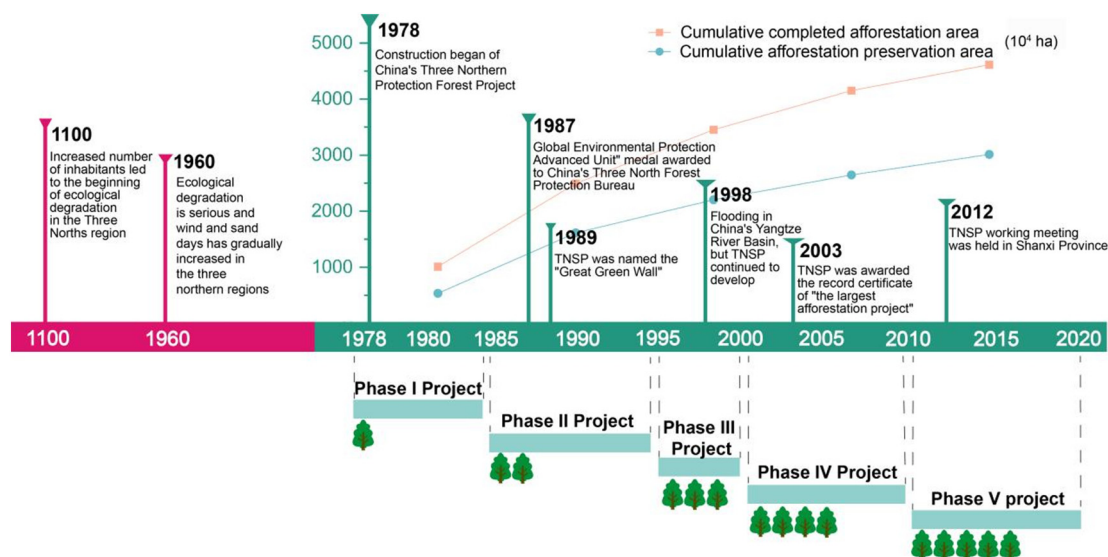


Fig. 1. Timeline of historical events in the Three-North region of China. Chronology of ecological conditions, historical events, and achievements received by the TNSFP in the Three-North region from 1100 to 2020. Also included are the trajectories of cumulative completed afforestation area (yellow) and cumulative afforestation preservation area (green) for Phase I to Phase V Programs. The TNSFP construction has significantly increased the total forest resources. Cumulative completed afforestation area (yellow) and Cumulative afforestation preservation area (green) are obtained from the Development Report on 40 Years of Construction of the Three North Protection Forest System (1978–2018). (Reference: (Administration SFAG, 2019)).

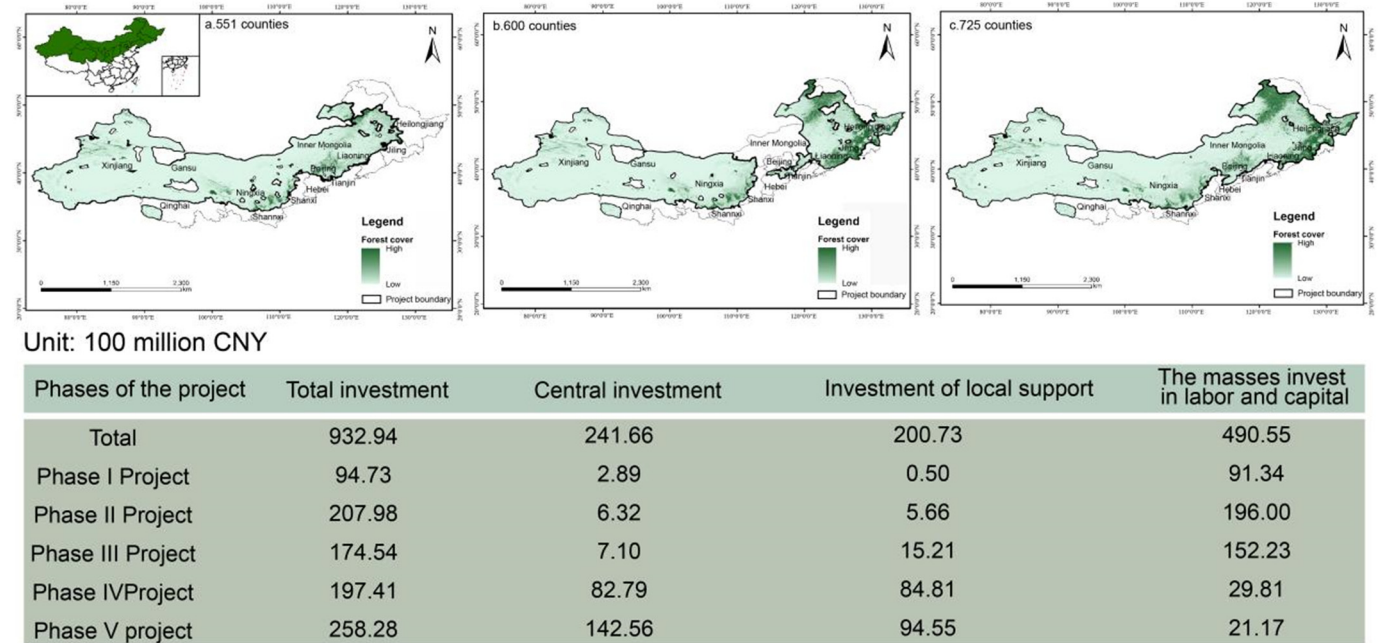


Fig. 2. Study area map of China's TNSFP. The maps include: the forest coverage and planning scope (a, b, c) of the third, fourth and fifth phases of the TNSFP; the Phase III planning area—551 counties (a); the Phase IV area—600 counties (b); and the Phase V area—725 counties (c). (Note: the Phase I Program planning scope was 406 counties, and the Phase II Program was 514 counties). The table shows the investment in each phase of the TNSFP from 1978 to 2017. The largest proportion (52.58 %) of the investments was in the labor capital. After entering the Phase IV Program, the state abolished the two worker categories (rural compulsory labor and accumulated labor), and the total amount invested in the labor capital dropped significantly. Tree cover data derived from the annual Vegetation Continuous Fields (VCF) tree cover product by NASA's MODIS. Investment form from the Development Report on 40 Years of Construction of the Three North Protection Forest System (1978–2018). (Reference: (Administration SFAG, 2019; Hansen et al., 2003)).

combating desertification and controlling dust storms, controlling soil erosion, enhancing forest carbon sequestration, increasing crop yield, and promoting economic development. We also particularly identified how the TNSFP was responding to and promoting the achievement of four relevant SDGs (SDG2, SDG8, SDG13, SDG15).

4.1. Combating desertification and controlling dust storms

Ecosystem service for wind and sand control is an important issue in the 43 years of TNSFP construction, and almost >1/3 of the literature focuses this issue. On this ecosystem service, the area of desertification and frequency of dust storms in four key wind and sand controlling zones has been significantly decreased in the past 43 years, namely, the Horqin Sands (Inner Mongolia Autonomous Region), the Maowusu Sands (Shaanxi), the Hulunbuir Sands (Inner Mongolia Autonomous Region), and the Hetao Plain (Inner Mongolia and Ningxia Autonomous Regions) (Chen et al., 2003; Wu and Ci, 2002). Afforestation can significantly increase the regional vegetation cover and reduce the area of desertification. For example, Yan et al. (2015) selected Landsat TM images (spatial resolution 30 m) of the Horqin sands in 2000 and 2010 for image analysis and path analysis, demonstrating that desertification in the horqin sands was controlled between 2000 and 2010, with a decrease in desertification area and intensity. Total vegetation cover increased by 10.6 % during the study period, with protective forests accounting for 12.8 % of the increase in total vegetation cover, 38.4 % of evapotranspiration, and 11.3 % of precipitation in the study area. Therefore, in addition to protective forests, climatic conditions such as evapotranspiration and precipitation can also indirectly influence the change of vegetation cover and desertification area (Li et al., 2009; Wang et al., 2010; Yan et al., 2011; Yan et al., 2015; Zhang et al., 2012).

We defined desertification intensity as slight (<5 % of total area by drift or wind erosion), moderate (5 %–25 %), severe (25 %–50 %), and extremely severe (>50 %), and estimated the area for each intensity level according to the method of Yan et al. (2009) and Zhu et al. (2017) (Yan et al.,

2009). From 2004 to 2014, the severe and extremely severe sandy land area in the Maowusu Sands decreased by 3.31×10^5 ha and 5.95×10^5 ha, respectively, while in the Hulunbuir Sands severe sandy and extremely severe sandy areas have been reduced by 4.78×10^4 ha and 2.02×10^4 ha, respectively. This reduction in the sandy land area is attributed entirely to the significant contribution of the TNSFP ecological restoration Program in the priority construction areas (Fig. 3a) (Administration SFAG, 2019; Feng et al., 2021; Xiu et al., 2018). We further compared the effectiveness of the major areas by land status, and found that the Maowusu sandy area was also reduced by up to 5.68×10^5 ha and 2.49×10^5 ha in the mobile and semi-fixed sandy area, respectively, which clearly showed the effectiveness of the TNSFP. In the Hetao Plain, the semi-fixed sandy land area increased by 9.38×10^4 ha (Fig. 3b) (Administration SFAG, 2019). The contribution of the TNSFP shelterbelt to the reduction in the desertification area was 14.9 %, which indicates that large-scale afforestation significantly reduced the frequency of dust storms in the Three-North region (Cao, 2008; Li et al., 2012; Piao et al., 2005; Wang et al., 2010; Xu et al., 2019; Zhu and Zheng, 2019).

Some studies question the positive effect of the TNSFP and the ability of afforestation to combat desertification and promote rehabilitation in the Three-North region of China (Wang et al., 2010). They do not believe that increased afforestation has contributed significantly to the long-term reduction of dust storms, despite previous claims that there was a causal relationship. The main reason for this is that in the Three-North region, the number of dust storms days has decreased continuously, even during periods when afforestation was not being carried out on a large-scale (Chen et al., 2003; Wang et al., 2010). The Hebei Dam Plateau and the Inner Mongolia Plateau have likewise not undergone large-scale reforestation, but recovery efforts from dust storms have continued during the same period. In the Taklamakan Desert in southern Xinjiang, the decreases in the frequency of dust storms do not seem to have resulted from the large-scale afforestation that has been extensively carried out by the TNSFP in the area since 1978 (Zhou and Zhang, 2003). This shows that at least in parts of the Three-North region, large-scale afforestation is not the key factor

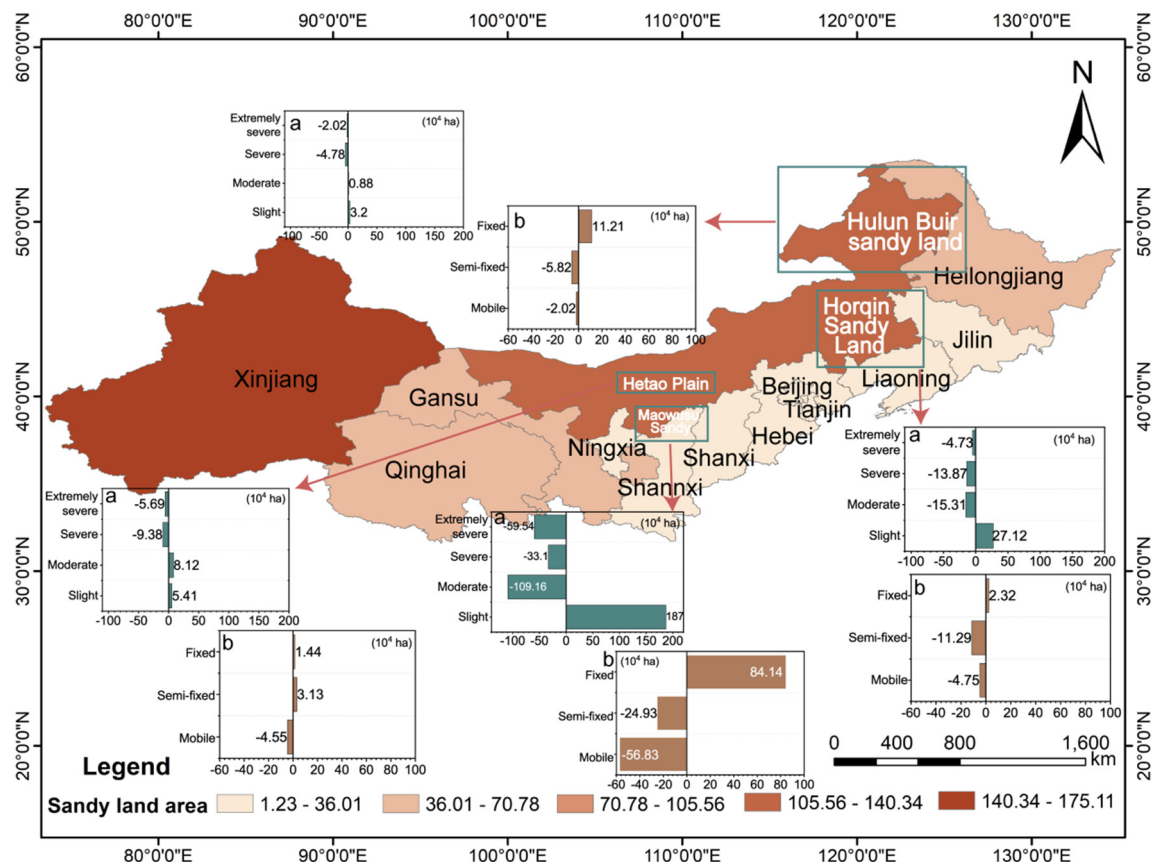


Fig. 3. The construction of protective forest system in the wind and sand areas of China's TNSFP (Horqin Sand, Maowusu Sand, Hulunbuir Sand and Hetao Plain). a. Key construction areas of wind and sand by severity level (slight, moderate, severe, extremely severe); b. Key construction areas of wind and sand by land status (mobile, semi-fixed, fixed). Map of construction by severity and land status from the Development Report on 40 Years of Construction of the Three North Protection Forest System (1978–2018). (Reference: (Administration SFAG, 2019)).

responsible for controlling desertification (Wang et al., 2010). Because land degradation caused by human activities such as overgrazing, land denudation, and overuse of groundwater resources offset the TNSFP afforestation efforts, thus diminishing the effectiveness of afforestation restoration efforts in reducing dust storms (Cao, 2011; Wu and Ci, 2002). The reduction of dust storms in the western and southern Taklamakan, and the Minfeng region of Xinjiang has been reported to closely relate with climate change (Wang et al., 2006; Zhou and Zhang, 2003).

4.2. Controlling soil erosion

In the Three-North region of China, soil erosion mainly occurs in the Loess Plateau region, which also has the largest area of soil erosion; approximately 90 % of the sediment in the Yellow River comes from soil erosion in the Loess Plateau (Chen et al., 2007; Li, 2000; Zhao et al., 2013). With the implementation of the TNSFP Program, the area of the Loess Plateau with a sediment yield $>5000 \text{ t}/(\text{km}^2 \cdot \text{a})$ decreased by 44.3 %, while the sediment load decreased significantly in the Yellow River and the intensity of soil erosion was substantially weakened (Jiang et al., 2019; Jiang et al., 2015; Yue et al., 2014). The sediment load decreased from $0.73 \pm 0.28 \text{ Gt}/\text{yr}$ during 1980–1999 to $0.32 \pm 0.24 \text{ Gt}/\text{yr}$ during 2000–2010, and the sediment discharge in 2013 was reduced to 0.2 Gt, which was similar to the level of the early agricultural period with less human activity (740 CE) (Chen et al., 2015; Wang et al., 2016). We compared river runoff from the TNSFP based on recent studies. We found that the TNSFP contributed a maximum of 67 % to the change in the water erosion area in the Yellow River basin, and the water erosion area in the Yellow River basin decreased from 29.8 % to 3.1 % (Fig. 4d). Moreover, the most significant decrease was observed in the Middle and Lower Wudinghe River and north-central Shanxi

Province, where sediment yield decreased by $>40 \%$, and the decrease was attributable to ecological restoration Programs mainly in the TNSFP (Li et al., 2021; Liang et al., 2015a; Liang et al., 2015b; Xin et al., 2009). The contribution of the TNSFP to the change of water erosion area in the Songhua River basin is at least 54 %, although the average multi-year runoff in the Songhua River basin was as high as $1.48 \times 10^{10} \text{ m}^3$ to $6.34 \times 10^{10} \text{ m}^3$ (Fig. 4d, a) (Administration SFAG, 2019).

We also found a positive effect of the TNSFP soil and water conservation forests on soil erosion control over a long time series. Soil erosion, broadly defined as “the wearing away of the land surface by physical forces such as rainfall, flowing water, wind, ice, temperature change, gravity or other natural or anthropogenic agents that abrade (Verheijen et al., 2009). Zhu et al. (2017) estimated the area and intensity of soil erosion in the three northern regions using the modified universal soil loss equation (RUSLE), and estimated soil erosion parameters in the study area based on local rainfall, topography, soil classification, and remote sensing data. Finally, the contribution of forest and shrub to the reduction of soil erosion was determined using through-path analysis. Studies have shown that the contribution of forests and shrublands in the TNSFP to the reduction in the soil erosion area was 37 % from 1978 to 1990, and its contribution increased to 40 % from 1990 to 2000 and 39 % from 2000 to 2008. In the last 30 years, the TNSFP forests and scrubs have contributed up to 53 % to the reduction of soil erosion areas. In the last 40 years (as of 2017), the increase in the TNSFP soil and water conservation forest area (forests and shrublands) contributed 61 % to the reduction of soil erosion, and the area of soil erosion in the Three-North region of China was reduced by 66.6 % compared with 1978 (Administration SFAG, 2019; Zhu et al., 2017). The TNSFP has steadily improved the key challenge of soil erosion in the Three-North region. In addition, the TNSFP showed different results

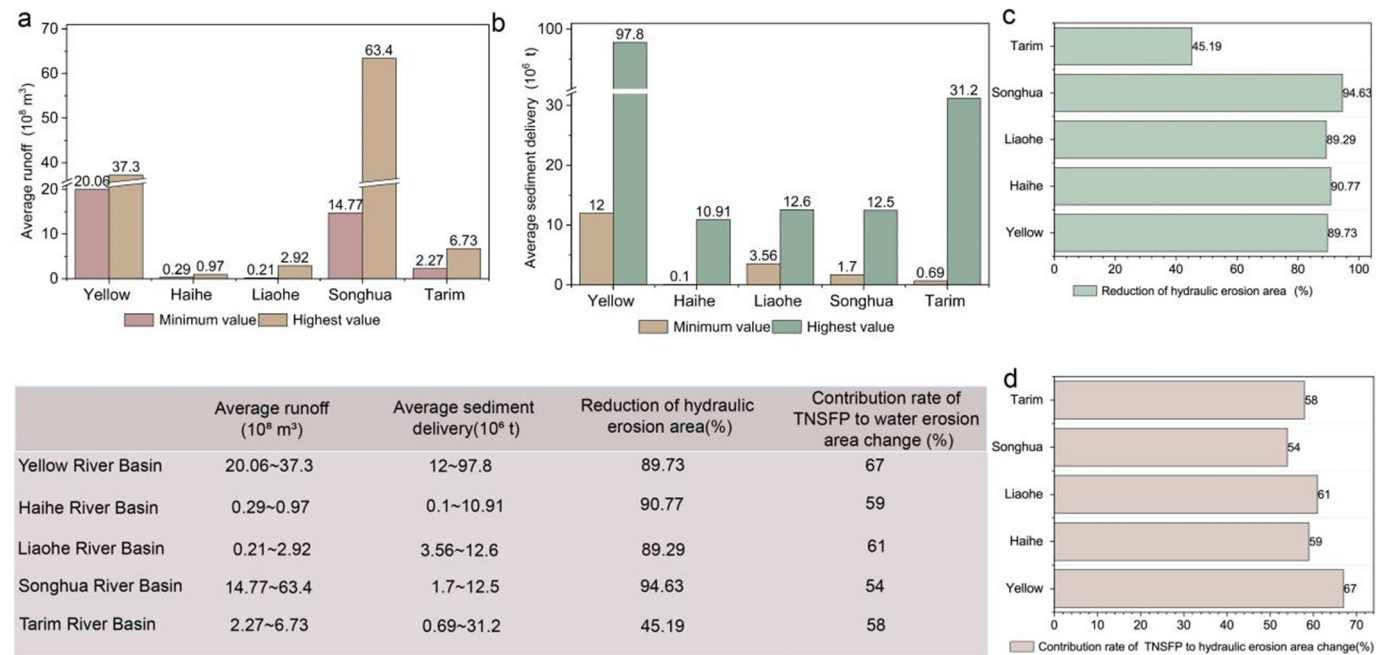


Fig. 4. Impacts of China's TNSFP on river runoff and sedimentation. The figure represents the average runoff (a), average sediment delivery (b), reduction of erosion area (c), and the contribution of the TNSFP to water erosion area change (d) in the Yellow River Basin, the Hai River Basin, the Liao River Basin and the Songhua River Basin. All data sources in the figure are from the Development Report on 40 Years of Construction of the Three North Protection Forest System (1978–2018). (Reference: (Administration SFAG, 2019)).

at different levels when controlling soil erosion. The area of very slight, slight, moderate, severe and extremely severe soil erosion decreased by 24 %, 34 %, 50 %, 83 % and 99 %, respectively, in the Three-North region of China (Zhu et al., 2017).

4.3. Enhancing forest carbon sequestration

The TNSFP has improved the forest carbon sequestration capacity, effectively relieved climate change, and had a positive impact on the carbon balance of China's terrestrial ecosystems (He et al., 2015; Ling et al., 2008; Liu et al., 2014; Piao et al., 2009; Zhang et al., 2021). It is reported in the literature that the total carbon sequestration increase of the TNSFP was 2.31×10^3 Tg C from 1978 to 2017, which corresponded to 5.23 % of China's industrial carbon emissions in the same period (1980–2015), with the highest percentage of over a 50 % increase in biological carbon sequestration from shelterbelts. The total carbon stock of the TNSFP was 124.3 Tg C from 2001 to 2010 (Administration SFAG, 2019; Lu et al., 2018). In spatial terms, the shelter forests in the central Loess Plateau have acted as a carbon sink, and carbon stocks in the scrub are all showing a trend of steady growth. Over the past 30 years (1978–2007), the carbon stock of the shelter forest on the Loess Plateau of the TNSFP amounted to 215.2 Tg C. The biomass density of the shelter forest on the Loess Plateau was nearly three times greater than that of other regions, and the carbon sequestration of the forest in other regions of the Three-North was lower (Administration SFAG, 2019; Feng et al., 2013; Liu et al., 2014; Lu et al., 2018; Sun et al., 2016). The TNSFP-based afforestation Programs have played an important role in increasing the carbon stock of forest ecosystems (Liu et al., 2014; Wu et al., 2018). Due to the incompleteness of the literature, we were unable to illustrate trends in the effects of TNSFP on forest carbon sequestration over a continuous long time series during our assessment.

Carbon stocks in the TNSFP differed significantly among forest types. Chu et al. (2019) used 1 km × 1 km gridded forest data in their study based on Landsat TM/ETM remote sensing images provided by the Resource and Environment Science Data Center of the Chinese Academy of Sciences (<http://www.resdc.cn>) in 1990, 1995, 2000, 2005, 2010, and 2015. Based on the INVEST model assessment, it was found that in 1990, total carbon sequestration was 6.97×10^5 Tg C in the TNSFP, with over two-thirds of

the carbon sequestration in mixed woodland (4.96×10^5 Tg C, 71.10 %). The contribution of closed shrublands and open shrublands to the total carbon sequestration of the TNSFP showed a decreasing trend, probably because of the conversion of forested landscapes to other landscapes, such as construction land or cultivated land, by human activities. In 2015, mixed woodland still accounted for the largest part of the total TNSFP carbon sequestration, contributing 4.83×10^5 Tg C (70.67 %) (Chu et al., 2019). The results of the TNSFP's carbon stock study in the Loess Plateau region support the conclusion that forest carbon sequestration capacity varies by forest landscape. For example, the researchers used data obtained from field surveys and Landsat ETM+ imagery to assess the carbon pools of protected forests in the central Loess Plateau of China. The study showed that in the TNSFP loess plateau area, the carbon stock of the protective forest was 215.2 Tg C from 1978 to 2007, and the tree forest accounted for 68 % of the total carbon stock, while the contribution of shrub woodland and herbaceous plants to the carbon sequestration of the TNSFP protective forest was extremely low (Sun et al., 2016).

There is a large gap in the study of carbon stock and its changes after the implementation of TNSFP. The spatial and long time series of data in the impact of TNSFP on carbon stock in the three northern regions of China and the reasons for carbon stock changes is lacked, although the work on carbon stock development and carbon neutrality is also at the forefront of future development. Therefore, an in-depth study of the carbon sequestration capacity of TNSFP has great potential in the future.

4.4. Increasing crop yield

The farmland shelterbelt area of the TNSFP increased by 1.34×10^6 ha and contributed to a cumulative increase in crop yield of 4.23×10^{11} kg in the TNSFP during 1978 to 2017. The contribution of shelterbelts to crop yield in the high, medium, and low climate potential productive zones classified by farmland was 13.3 %, 17.7 % and 10.9 %, respectively (Administration SFAG, 2019; Zhu and Zheng, 2019; Zhu et al., 2017). Sun et al. (2010) sampled wheat yields by multiples of the height of the trees in the protective forest through field surveys, and then used the sample yields to obtain wheat field yields by converting them to yield increase = $[(W_1 - W_2) / W_2] \times 100$ % analysis (W_1 yield in yield increase area; W_2

control yield) with statistical analysis, and concluded that Wheat yields were significantly higher in farmland with protective forests compared with farmland without protective forests in Northwest Gansu Province (Sun et al., 2010).

The impacts of the TNSFP on local crop yield in the Three-North region are mainly focused on the northeastern region (Chen et al., 2013; Zheng et al., 2016). The grain yield increase in the Northeast Songnen Plain was 3.51 %, while the grain yield in the Songliao Plain increased by 53.62 % per unit (Administration SFAG, 2019). In a previous study, Zheng et al. (2016) estimated the crop (i.e., maize) yield in Northeast China using the harvest index method and MODIS data and calculated the level of protection of farmland by shelterbelts in the study area using Landsat images of time series. The results of this study showed that shelterbelts can increase crop yields on a regional scale, with grain yield increases of 4.7 %, 4.3 %, and 9.5 % in areas of high, medium, and low different production potential in Northeast China, respectively. Farmland shelterbelts are more effective in increasing yields in low-production areas. This conclusion is also supported by Jilin farmland protection forests. Shi et al. (2016) conducted a study on how Jilin farmland protection forests affect crop yields based on remote sensing (RS) and geographic information system (GIS) technologies, and concluded that Jilin farmland shelterbelts had a more significant effect on increasing yields in low-production areas where natural conditions were poor, with a yield increase of 8.85 % (Shi et al., 2011; Shi et al., 2016; Zheng et al., 2016). In summary, the areas with poor natural ecological conditions have poor land productivity, so the rate of yield increase keeps increasing with the establishment of protective forests. However, the yield increase rate declines as the natural conditions become better, so the protection effect is not as obvious as before. However, when the protective effect of the shelterbelt was above 80 %, the farmland areas neighboring the shelterbelt usually showed lower food productivity owing to shading effects and competition with the shelterbelt for water, light and nutrients (Ding and Su, 2010; Rivest and Vézina, 2015).

However, there are many uncertainties about the impact of TNSFP shelterbelts on food production. For example, the features of the shelterbelts themselves, climate change and social shocks (abrupt harvests, market prices and input costs) may affect grain production (Chappell et al., 2013; Chuku and Okoye, 2009; Kort, 1988; Zheng et al., 2016). The positive impact of shelterbelts on grain production decreases as extreme weather such as drought occurs or when precipitation increases (Easterling et al., 1997; Leng and Huang, 2017). Therefore, in the Three-North region of China, we need to have a comprehensive understanding of the link between climate and grain yield. This will increase our understanding of the adaptive capacity of grain to the effects of climate change and various shocks to better leverage the positive effects of TNSFP shelterbelts on grain yield (Leng and Huang, 2017; Wheeler and Von Braun, 2013).

4.5. Promoting economic development

Available evidence suggests that the TNSFP has contributed to the economic development of the Three-North region through a multifaceted and diverse approach (Cao et al., 2020). Over the past 43 years, the TNSFP construction has vigorously developed ecological economic forests with apples, peaches, red dates, chestnuts, jujubes, apricots and other special varieties, and built high-quality apple bases mainly in Shanxi, Shaanxi and Gansu, with some key areas where forest and fruit income has accounted for >50 % of farmers' net income. The production of some economic forest products in the TNSFP area Shaanxi apples (7.60×10^9 kg), Hebei peaches (2.09×10^9 kg) and chestnuts (3.81×10^8 kg) and Xinjiang apricots (9.30×10^8 kg) and dates (2.75×10^9 kg) all accounted for the largest proportion in the Three-North region (Fig. 5a) (Administration SFAG, 2019). The TNSFP absorbed a total of 3.13×10^8 surplus farm laborers during the construction period, among which 1.00×10^7 rural laborers were absorbed by afforestation, contributing 27 % to local poverty alleviation (Administration SFAG, 2019; Li et al., 2012; Li et al., 2021). The TNSFP has increased farmers' income, promoted poverty alleviation, and improved agricultural productivity. Farmers' income in the Loess Plateau region

increased from 1916 CNY in 2000 to 11,224 CNY in 2017, with an annual growth rate of 11.04 %. The above multifaceted local economic development is attributed to the changes in TNSFP construction (Administration SFAG, 2019; Li et al., 2012; Li et al., 2021).

With the improvement of the ecological environment in the TNSFP area, ecotourism in the local engineering area is developing rapidly. From 2004 to 2016, a total of 8572 forest parks were built in the region, including 3615 at the national level and 4357 at the provincial level, and the improved environment has stimulated the development of tourism (Administration SFAG, 2019). As of 2018, the Three-North region received 384 million visitors, with direct tourism revenue of CNY 48 billion, providing employment opportunities for residents. Liaoning Province accounted for >30 % of the revenue (14.8 billion CNY) (Fig. 5b, c) (Administration SFAG, 2019). The TNSFP promoted the combined development of ecotourism, red tourism, and folk tourism. For example, Nanniwan village in Yan'an city has taken full advantage of its red tourism resources (history and cultural heritage) to create featured town tourism incorporating communist sites, ecological landscapes, and leisure agriculture. This town welcomed approximately 1.30 million tourists in 2017, which yielded 25 million CNY in tourism income and contributed to the increase of per capita net income by 3468 CNY in 2017 (Li et al., 2021). In general, TNSFP construction combines forest promotion and enrichment, and the development of economic benefits in all aspects of the project area cannot be separated from TNSFP construction. While reviewing this ecosystem service, we have indirectly shown that TNSFP does promote economic improvement in terms of forest fruits, farm income, and tourism, while there are gaps in the direct economic changes brought about by the construction of the project, and we seriously lack direct evidence.

4.6. Contributing to the Sustainable Development Goals

The UN 2030 Agenda for Sustainable Development and its 17 SDGs serves as the overarching framework for sustainable development, calling for universal and inclusive action to address the multiple and complex challenges that the world is facing in the 21st century (FAO Food and Agriculture Organization of the United Nations, 2018). We analyzed the ecological and economic benefits by the TNSFP for the program area, further identified the status of sustainable development in the study area and provided a basis for clarifying the key tasks for the future sustainable development of TNSFP by clarifying it in the context of SDGs.

In the case of Zero Hunger (SDG2), hunger and malnutrition are barriers to sustainable development. With the impact of the TNSFP program, food production has increased and hunger in the study area has gradually relieved, but we did not find any relevant literature on nutritional status. In decent work and economic development (SDG8), increasing employment opportunities and promoting sustainable economic growth are considered as a major challenge. The TNSFP can absorb the surplus labor of local farmers during its construction, and the level of labor security and the employment rate has also been improved. Climate Action (SDG13), the TNSFP construction enhances forest carbon sequestration, but extreme weather events from climate change are uncontrollable. For terrestrial biology (SDG15), combating desertification and controlling land degradation, the TNSFP is effective. For more sustainable development of the three northern regions, there is still a need to strengthen sustainable ecosystem management. Overall, the TNSFP as the world's largest afforestation Program focuses on afforestation and reforestation, however, a range of other factors respond significantly and synergistically to this process. The TNSFP plays a nonnegligible role in driving SDGs.

5. Lessons from the TNSFP

The TNSFP has made significant progress in promoting ecosystem services and economic benefits. However, the TNSFP still has problems with afforestation. Over the past 40 years, the TNSFP had relatively low silvicultural preservation rates and poor quality protective forests with high risk of decline (Administration SFAG, 2019). The main reason is that >90 % of

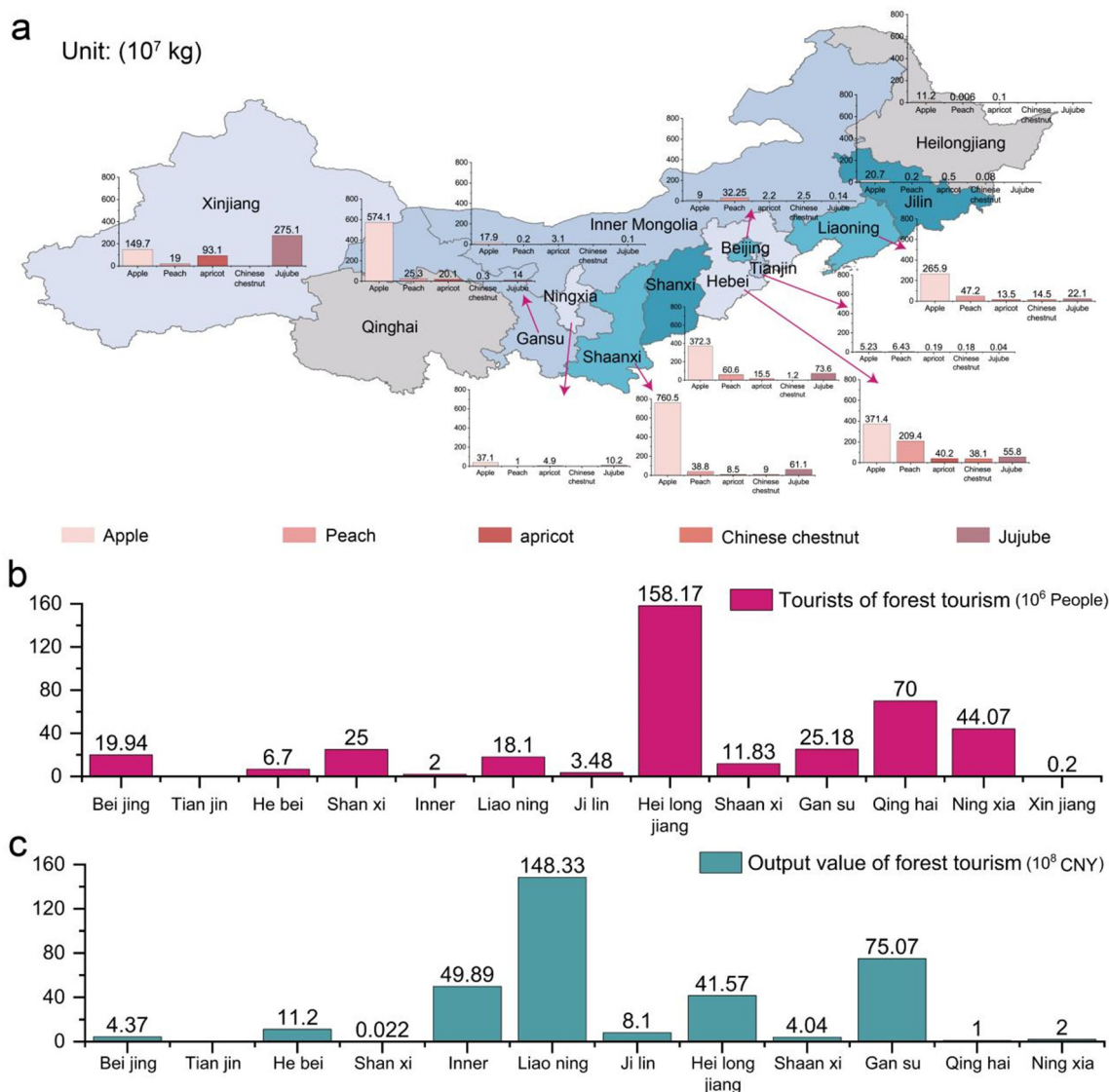


Fig. 5. Changes in forestry economic output in different study regions of the TNSFP. a. Spatial distribution statistics of production of a range of economic forestry products in the TNSFP region. b. Statistics on the number of visitors received by the TNSFP region. c. Forest tourism output in the TNSFP region. All data sources in the figure are from the Development Report on 40 Years of Construction of the Three North Protection Forest System (1978–2018). (Reference: (Administration SFAG, 2019)). (Note: Some regions lack data).

northwest China is desert grassland and desert, which is prone to low survival rate of forest trees without proper assessment of the natural conditions of the afforestation zone. One obvious lesson learned from this assessment is that the desertification control effect of protective forests is much lower than expected, and governments must fully consider the choice of water resources and tree species when implementing mechanisms, in addition to raising the ecological awareness of the population is crucial (Song et al., 2014; Zheng et al., 2012; Zhu et al., 2017). Previous studies have also shown that the TNSFP farmland protection forests are poorly protected, with the actual degree of protection effect being only 10.1 % (Zheng et al., 2016). The ratio of trees to irrigation in arid and semi-arid areas is unreasonable, and the proportion of arboreal forests is large. The ecosystem remains fragile and the expansion of vegetation greatly affects the actual evapotranspiration, leading to moisture reduction and soil drying (Cao, 2011). Based on these lessons, we have identified three pressing challenges that governments, scientists, policymakers, and other stakeholders must address as they embark on building the TNSFP's next decades of conservation.

6. Challenges and suggestions for the TNSFP

Over the past 40 years, the TNSFP has made substantial contributions to combating desertification and controlling dust storms, controlling soil erosion, enhancing forest carbon sequestration, increasing crop yield, and promoting economic development. However, the TNSFP still faces significant challenges. Based on our comprehensive assessment, we found that the construction of the TNSFP still has gaps in its scientific basis, limited local participation, and insufficient investment efforts, and these challenges are intertwined. Here, we try to describe these challenges in detail and provide scientific guidance for the long-term sustainability and future direction of the TNSFP.

6.1. Expanding the scientific evidence

Based on the present evidence, few studies on the dynamics of the TNSFP in space and time have been documented and research gaps are still existed. For example, there is limited analysis on the regularity for

long time pattern of the wind–sand priority construction areas; the research on soil erosion and grain yield has focused more on the Loess Plateau and the Northeast, and there is little evidence of the effect of the TNSFP contribution in other parts of the country. This makes it difficult to fully assess which ecosystem services are most vulnerable to such dynamics, or how these changes affect these five ecosystem services. In addition, we found that there was limited in our knowledge when it came to understanding the silvicultural contributions of the TNSFP in an interdisciplinary and multifaceted manner. Most studies measuring the success of the TNSFP have focused on wind and sand control, without fully considering the multiple contributions of the TNSFP, especially the response to SDGs.

The timely expansion of the scientific knowledge base is key to maximizing understanding of the contribution of the TNSFP. More generally, analysis in both temporal and spatial dimensions is essential to understanding the environmental, economic, and social impacts of the TNSFP. We need to discuss the effects of the TNSFP construction using the most comprehensive evidence. In the process of expanding the scientific basis, the implementation of both short-term and long-term monitoring approaches is critical to the future success of recovery in China's Three-North region. We need to clarify the process of the TNSFP effects on the five indicators shown above, including the kind of changes that occurred, the timing and duration of change, and the magnitude of change. When spatial dimension analysis is performed, it should consider factors from the micro-scale to the macro-scale and from the household level to the national level, such as in the study of key construction areas versus the entire Three-North region (Fig. 6) (Qiu et al., 2017; Xi et al., 2012; Xi et al., 2014). Further enhancing data transparency would help improve the dynamic research and reporting of the TNSFP, along with the need to better incentivize and track continuous changes in TNSFP construction. We emphasize that holistic, interdisciplinary, and multifaceted research facilitated our enhanced understanding of the TNSFP, and it has brought much additional knowledge to the comprehensive analysis of the TNSFP contribution, including research gaps in the five indicators. Thus, we can also attempt to quantify the contribution of the TNSFP in terms of education and consumption and production.

6.2. Strengthening local engagement

There are substantial risks for sustainable ecological restoration through the TNSFP. The mismatch between afforestation species and site conditions in the TNSFP can result in competition between the shelter forest and adjacent farmland areas. Reduced compensation has driven herders to overgraze and reclaim the land, which has led to increased desertification

(Cao et al., 2011; Wu and Ci, 2002). All the issues are attributed to the current challenges faced by the Program, with insufficient local participation. As the afforestation policy is implemented, the risk may decrease. These issues may not be fully resolved soon, but risk reduction must become a top priority for the government.

Strengthening local engagement is also important to ensure that China's sustainability interventions are continued (Fig. 6) (Bryan et al., 2018). From the experience and lessons learned from the TNSFP construction, we propose some strategies to mitigate these risks in the face of uncertainty. First, we suggest combining local participation in policy formulation and implementation, considering local environmental, economic, and social changes, and combining the selection of local regional tree species to increase total quantity and improve quality. According to the current plan, about 3.33×10^6 ha of forest and grass vegetation will be planted each year. Second, in addition to strengthening forest management, enhancing forest conservation, and improving forest quality, the issue of stand rehabilitation and restoration, including the rehabilitation of low quality and low efficiency forests, also requires local participation in management (Sudmeyer and Flugge, 2005). Improving local participation also requires that existing vegetation, especially natural vegetation, to be protected effectively. The destruction of vegetation and ecology should be severely combated and further strengthened. Finally, promote people's livelihood. For example, local governments should compensate local herders in the Three-North region for sustainable forest management to ensure that livelihoods are not lost where grassland is converted to forest. With appropriate compensation, herders can be involved in the development of economically beneficial and ecologically sustainable livestock management strategies. Herders should also be provided with a more comprehensive understanding of forest and grassland degradation to achieve adaptive responses and reduce overgrazing. In addition, local institutions are needed to provide feedback on the implementation of the TNSFP to regional and national governments during the implementation process.

6.3. Secure optimal investment

In terms of funding, TNSFP-based investments in ecological Programs cover only one-third to one-half of the provinces, and annual and target investments in different provinces vary widely (Yu et al., 2022). For example, North China and Northwest China received investments of 2.9×10^{10} CNY and 2.86×10^{11} CNY, while the Northeast only received \$3.260 billion. Furthermore, the TNSFP's Inner Mongolia region experienced a one-year break in the construction of protective forests (2005), and the Tianjin

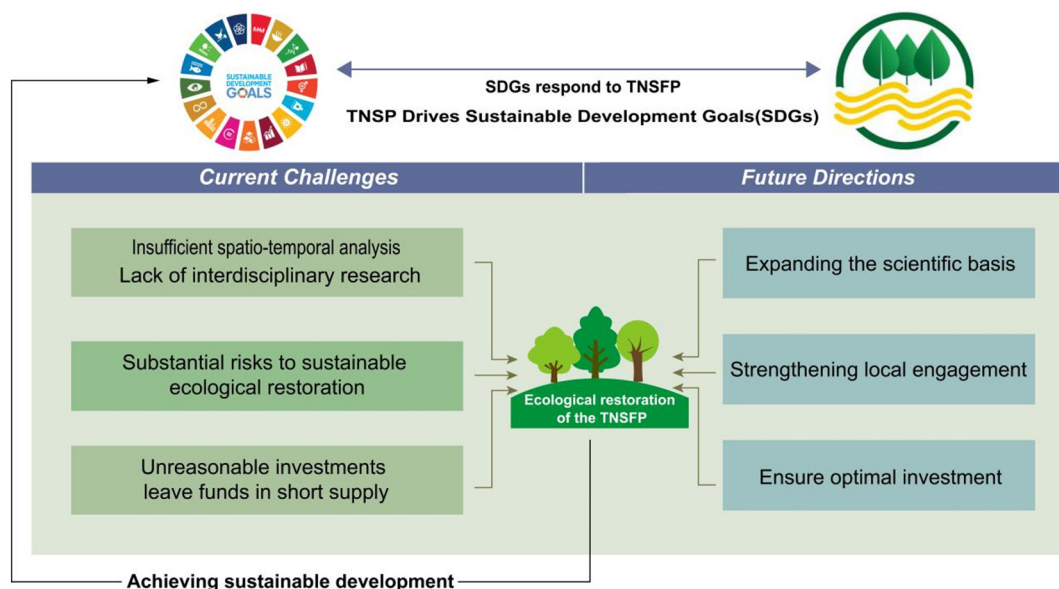


Fig. 6. Challenges and future directions of China's TNSFP.

region experienced short-term breaks in 2017 and 2018 (Yu et al., 2022). In terms of indicator investment differences, we found that the response benefits of combating desertification and controlling dust storms, as well as controlling soil erosion from the TNSFP were significant, with 14.9 % and 61 % contribution from protective forests, respectively, probably because of the considerable investment (Administration SFAG, 2019; Li et al., 2012; Wang et al., 2010; Zhu and Zheng, 2019). However, the uneven investment approach means that eco-investment still faces a problem of underfunding (Fig. 6).

In-depth understanding of the current situation and characteristics of ecological investment, to ensure adequate and optimal investment, and constantly improve the investment and financing mechanism for the construction of the TNSFP is essential for sustainable ecological development. In ensuring optimal investment, we recommend the allocation of more adequate financial resources and a more rational spatial distribution to maximize the results of the TNSFP program. Therefore, we need to improve the TNSFP investment mechanism. The establishment of government input should be the main channel, vigorously absorbs social resources and forces to participate in the TNSFP construction investment system. Actively guide private capital and various public welfare funds to invest in the field of TNSFP forestry and grass ecological construction and protection and utilization. Strive for a more sustainable and stable way of operation and management. We also believe that the implementation of national financial support policies for forestry, the successful experience of “purchase afforestation” in Shanxi and the practice of “accountable management” in other forestry Programs, exploring and experimenting with cooperative afforestation, share afforestation, contract afforestation and other forms, giving full play to the enthusiasm of local farmers to participate in the construction of TNSFP Programs, will greatly enhance the ecological services of the TNSFP. Finally, we suggest strengthening international cooperation in the field of ecological construction, broadening funding channels, attracting social and financial capital to enter the field of TNSFP forestry and grass ecological construction and protection and utilization, insisting on the combination of “bringing in” and “going out”, actively and deeply participating in global ecological governance, and extensively seeking extra-territorial funding. As China continues to accelerate investment in ecologically sustainable development under the 2030 Agenda for Sustainable Development, innovative financing mechanisms should be established and implemented to ensure adequate, sustained and stable investment to meet current ecological challenges (Yu et al., 2022).

7. Conclusions

The TNSFP, the world's largest and most representative forestry ecological Program to date, promotes the synergistic and comprehensive development of the environment, economy, and society. We synthesized >100 peer-reviewed articles on ecological restoration during 43 years of TNSFP construction and found that: (1) Combating desertification and controlling dust storms, controlling soil erosion, enhancing forest carbon sequestration, increasing crop yield, and promoting economic development are the more critical research hotspots, and all five aspects have achieved significant successes. (2) In terms of SDGs, the TNSFP has made great efforts for addressing zero hunger (SDG2), economic growth (SDG8), climate change (SDG13), and terrestrial biology (SDG15). (3) Addition to the successful experience of the TNSFP, there are two key problems with the Program in terms of afforestation, limitation of water resources and the selection of appropriate tree species. (4) Three great challenges of the TNSFP cannot be neglected; lacking scientific research evidence limits better public understanding of TNSFP construction; limited local participation hinders the full implementation of the policy; and irrational investment results in insufficient funding and unsustainable ecological restoration. (5) Three specific recommendations are proposed in response to the lessons and challenges: expanding the scientific basis, promoting local participation, and ensuring adequate and optimal investment. Our assessment provides an empirical basis for the future construction of the TNSFP. The TNSFP construction experience also forms a reference for other ecological engineering

constructions in China and elsewhere to ensure the maximum benefit of forest construction for global ecological security and the sustainable development of human society.

CRedit authorship contribution statement

Junjie Zhai: Conceptualization, Methodology, Writing, reviewing and editing. **Ling Wang:** Data collection and analysis, Writing. **Yuan Liu:** Data collection and analysis, Writing, reviewing. **Chengyuan Wang:** Data collection. **Xuegang Mao:** Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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